Curriculum Intelligence Engine

Overlap, redundancy, missing prerequisites and industry gaps in NMIMS B.Tech syllabi

Group 9 · J048, J049, J050, J052, J053 · AI University NLP Course Project 2026

The committee's three questions

1

Which courses teach the same thing?

Overlap and redundancy between courses, programmes and years

2

Which prerequisites are missing?

A declared prerequisite that no earlier semester teaches

3

What does industry ask that we do not teach?

Skills in job descriptions compared with what the syllabi cover

Output: structured JSON with evidence and a confidence score, for a human decision

Data: NMIMS semester syllabus booklets

2,413

non-exam PDFs in the archive

531

with a text layer, parsed (no OCR)

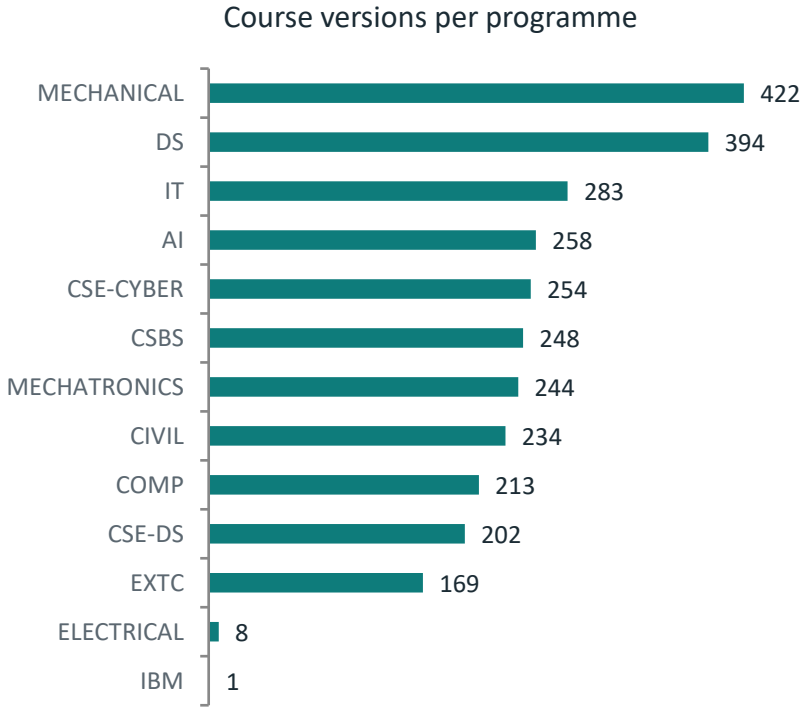
2,366

course versions after deduplication

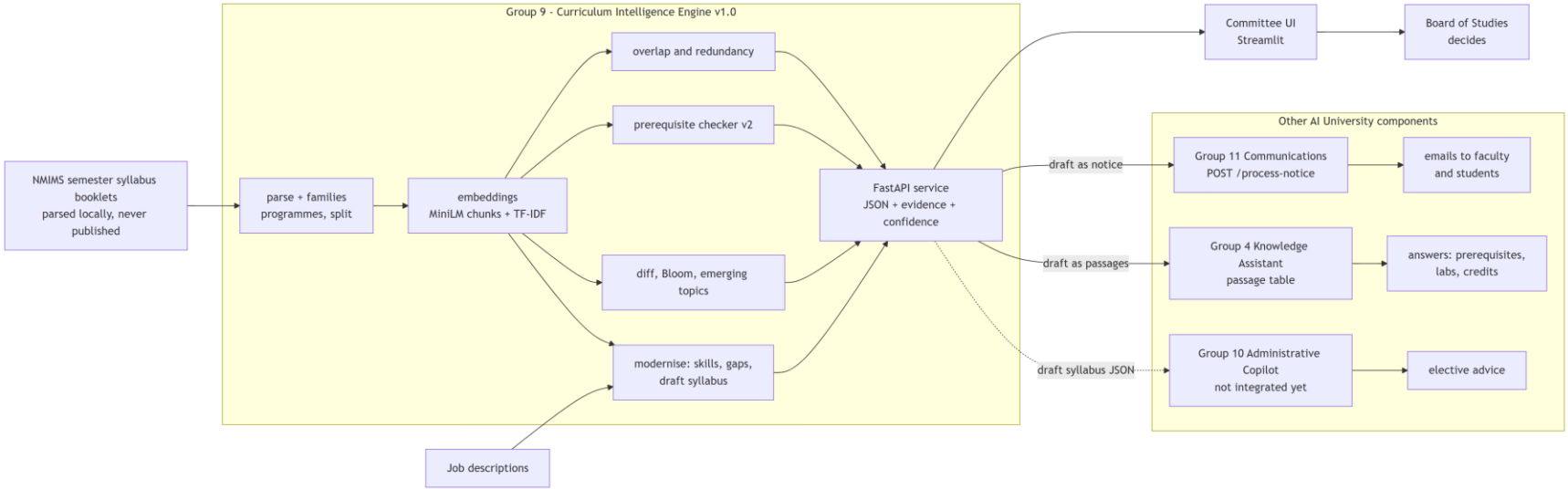
645

course families, 13 programmes

Academic years 2020-21 to 2026-27 · split by course family 60/20/20 (hash 4763325fd5)



Architecture



Runs offline

no cloud services; data never on GitHub

Evidence with every answer

matched units, sources, confidence

9 API endpoints

one JSON for the UI and peers

How we measured quality

502

LLM-annotated reference labels

35

settled by a third pass, 1 left uncertain

Label set	Items	Passes 1 and 2 agree	Settled by pass 3
course pairs (overlap 0/1/2)	200	171	28
detector flags (Week 3)	187	182	5
prerequisite v2 flags (Week 6)	45	45	0
JD skill coverage (Week 7)	70	68	2

No human annotators were used; results are indicative.

Second labeller: Phi-3 for part of the pairs, otherwise a second Claude prompt, so agreement there is not independent.

Results against our success criteria

12 / 14

success targets met on the test split

2

missed: both on prerequisite flags

Capability	Metric	Final	Target	Met?
Overlap	Spearman	0.717	≥ 0.6	yes
Redundancy	precision of top flags (reference labels)	0.800	≥ 0.7	yes
Prerequisites	removal recall	0.711	≥ 0.8	no
Prerequisites	precision of flags (reference labels)	0.500	≥ 0.6	no
Curriculum diff	precision (real year pairs)	0.998	≥ 0.85	yes
Bloom	accuracy vs faculty tags	0.776	≥ 0.7	yes
Emerging topics	precision of flagged gaps	0.915	≥ 0.7	yes
Topic labels	share accepted	0.700	≥ 0.7	yes
Curriculum modernisation	taught / gap accuracy on JD skill themes (reference labels)	0.857	no target set	–

Overlap and redundancy work

0.717

Spearman with reference labels (target 0.6)

0.965

ROC-AUC, any overlap (target 0.85)

0.800

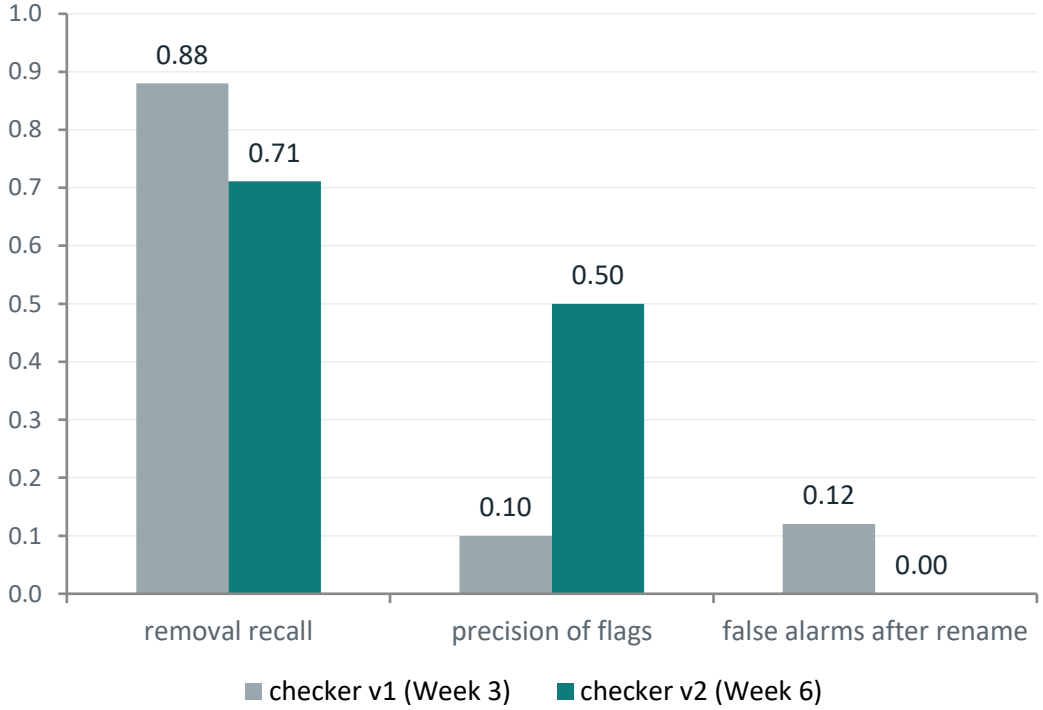
precision of top redundancy flags (target 0.7)

e.g.

3 of 10 NLP-course pairs are identical

Natural Language Processing is taught with the same units in AI, COMP / CSE-DS / IT and EXTC; Speech and NLP overlaps it strongly; Generative AI hardly overlaps. A committee can merge these into one course.

Prerequisites: an honest trade-off



Fewer false alarms

rename false alarms 0.120 → 0.000

Higher precision

0.100 → 0.500 (n = 14); target 0.6 still missed

Lower recall

0.880 → 0.711; below the 0.8 target

Demo: from JDs to a new syllabus

1 Input

5 JDs + 5 current NLP syllabi

2 Overlap

3 of 10 syllabus pairs identical

3 Skills

15 demanded: 6 taught, 9 gaps

4 Draft

11 units: 5 shared, 2 kept, 4 new

Group 11 · Communications

Notice in their /process-notice schema → one email per audience (this run: mock; live once their service runs)

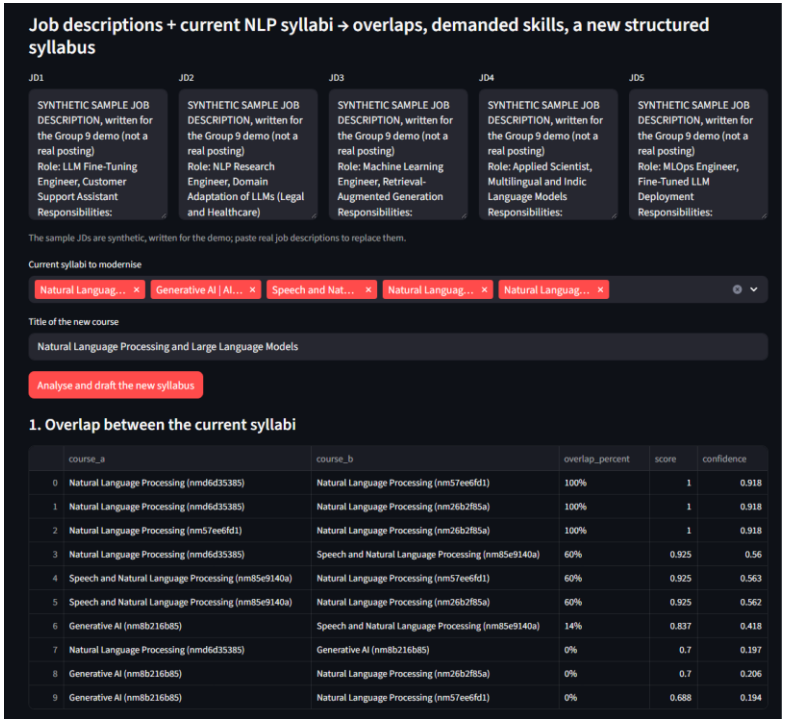
Group 4 · Knowledge Assistant

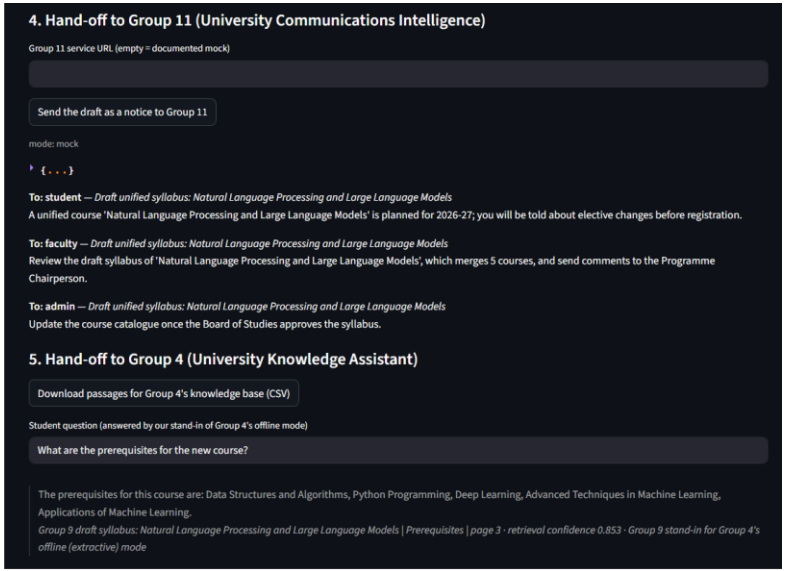
17 passages in their schema; our stand-in answered 4 of 5 student questions

Group 10 · Admin Copilot

Can read the same draft syllabus JSON for elective advice (not integrated yet)

Committee UI





Left: JDs, selected syllabi, overlaps. Right: Group 11 emails (marked mock), a Group 4 answer with its source.

Responsible AI

1 Privacy

Booklets and parsed text stay local; only ids, metrics and labels are public

2 No invented content

Draft syllabi are copied or templated, never generated; every unit names its source

3 Human in the loop

Outputs are drafts for the Board of Studies; the UI warns against automatic course cuts

4 Fairness

Bloom accuracy ranges 0.692 (COMP) to 0.857 (MECHATRONICS); small programmes excluded from prerequisite checks

Limitations and next steps

Limitation	Today	Next step
Scanned booklets	531 of 2,413 PDFs parsed	Add OCR, then re-run coverage
Labels	LLM-annotated, no human annotators	Fill the 50-pair human spot check
Prerequisite flags	precision 0.500, removal recall 0.711	Faculty-confirmed prerequisite map
Job descriptions	synthetic samples in the demo	Real postings; Group 10 integration

Every number in the reports and these slides is generated from results/.json; skipped analyses are reported as not run.*

Thank you · Questions?

github.com/jainamsethia/Curriculum-Intelligence-Engine (tag v1.0)

36 tests pass; a clean clone reproduces all 62 result files

Code and text written with AI assistance (Claude); reference labels are LLM-annotated.